

WMDR2-core

Package in package 'Model'

WMDR2-core
Version 0.2.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-22

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core v0.2.x

v0.2.0

- time (a temporal extent) shall be realized as time:interval (unchanged from v.0.1.13)
- Following the OGC Features model, geometry = geolocation = coordinates + elevation
- All temporal<Name> elements are renamed to historical<Name>.
- historical<Name> elements other than historicalLocation shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,

```
historicalTerritory = [{  
  "date": "2000-01-01",  
  "territory": "<ISO1>"  
},  
{  
  "date": "2024-07-13",  
  "territory": "<ISO2>"  
}]
```

- "observation" describes time-series (collections) of (individual) observations / measurements.
- "observation"s are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- "deployment" is removed as a concept, most attributes are directly under "observation". - "instrument" links directly to "observation". In alignment with OMS/STA, there is a link to "facility" via "deplyoment", but this is not expected to be used at this point.
- "observation"s can be grouped to an "observationSets" (NB: This is a test and not guaranteed to exist in future.)

WMDR2-core
Version 0.2.0
jkl created on 2026-03-06. Last modified 2026-06-22

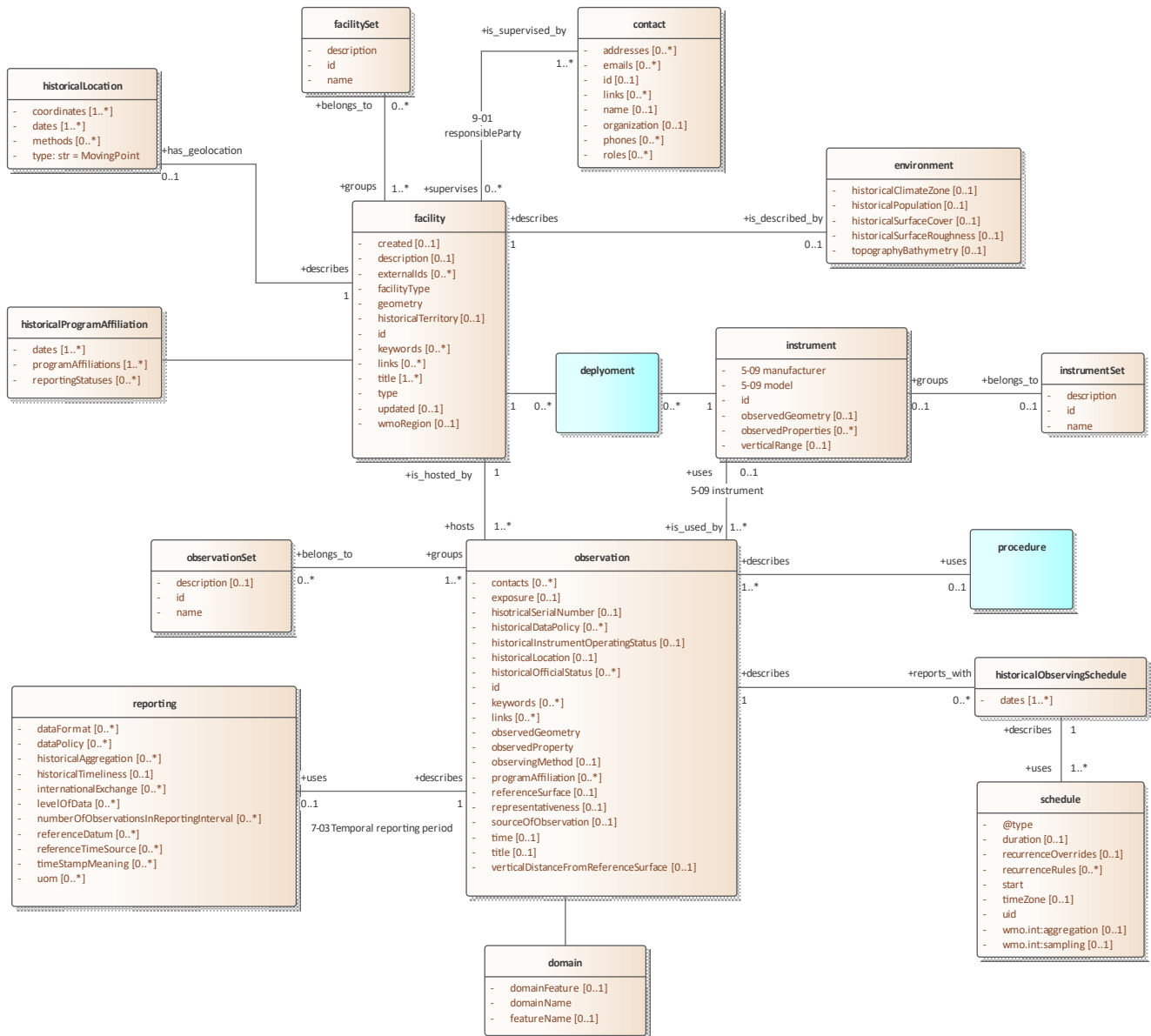


Figure 1: WMDR2-core

WMDR2-core-logical diagram

Class diagram in package 'WMDR2-core'

WMDR2core

Notes

1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, one for the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].
3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

version 0.1.4: simplify observation, specifically specification of 1-01

WMDR2-core-logical

Version 0.1.4

Version 0.1.4
jkl created on 2026-04-13. Last modified 2026-04-13

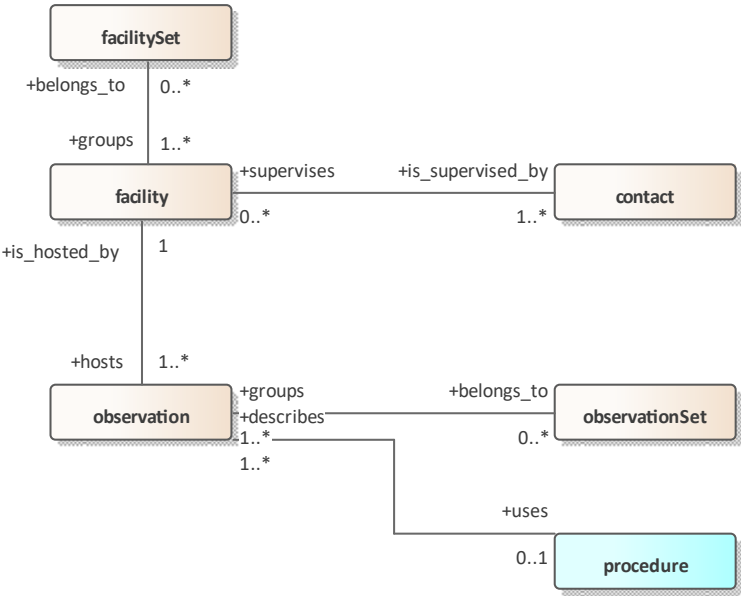


Figure 2: WMDR2-core-logical

contact

Class in package 'WMDR2-core'

WMDR2 core JSON class: contact.

contact
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-02

ATTRIBUTES	
addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Postal addresses. [Is static True. Containment is Not Specified.]	
emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts. [Is static True. Containment is Not Specified.]	
id : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact.	

ATTRIBUTES	
[Is static True. Containment is Not Specified.]	
 name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard)	[Is static True. Containment is Not Specified.]
 organization : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss	[Is static True. Containment is Not Specified.]
 phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts.	[Is static True. Containment is Not Specified.]
 roles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Specific contact roles such as owner, pointOfContact, principallInvestigator.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) 9-01 responsibleParty	
Source: Public supervises (Class) facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]

deployment

Class in package 'WMDR2-core'

deployment
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-22. Last modified 2026-06-22

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) deployment Cardinality: [0..*]	Target: Public (Class) instrument Cardinality: [1]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) facility
Cardinality: [1]

Target: Public (Class) deployment
Cardinality: [0..*]

domain

Class in package 'WMDR2-core'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

domain

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-10. Last modified 2026-06-18

ATTRIBUTES

 domainFeature : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation.

Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like

- river basin
- lake
- river
- catchment

- canopy

- epilimnion
- hypolimnion

- troposphere
- tropopause
- stratosphere

- thermocline (useful as a general term, but needs "domain")
- boundary layer (useful as a general term, but needs "domain")

- ...

[Is static True. Containment is Not Specified.]

 domainName : Private

The largest scale in WIGOS, cf. <http://codes.wmo.int/wmdr/Domain>

[Is static True. Containment is Not Specified.]

 featureName : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin"

[Is static True. Containment is Not Specified.]

ASSOCIATIONS



Association (direction: Unspecified)

Source: Public (Class) observation

Target: Public (Class) domain

environment

Class in package 'WMDR2-core'

Optional facility-environment object containing temporal climate, surface-cover, population, and topography/bathymetry histories.

environment

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-02

ATTRIBUTES



historicalClimateZone : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Temporal climate-zone history.

[Is static True. Containment is Not Specified.]



historicalPopulation : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]



historicalSurfaceCover : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Temporal surface-cover history.

[Is static True. Containment is Not Specified.]



historicalSurfaceRoughness : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]



topographyBathymetry : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

ASSOCIATIONS



Association (direction: Unspecified)

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public is _described_by (Class) environment
Cardinality: [0..1]

facility

Class in package 'WMDR2-core'

Facility properties payload of the WMDR2 full record. Root Feature geometry/time are modeled on fullRecord, not here.

facility
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES	
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record creation date/time metadata.	[Is static True. Containment is Not Specified.]
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Facility description.	[Is static True. Containment is Not Specified.]
 externalIds : Private Alias: 3-06 (primary & additional) WSIs Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) External identifiers such as WMO:WIGOS.	[Is static True. Containment is Not Specified.]
 facilityType : Private Alias: 3-04 facilityType Compact WMDR facility type code value.	[Is static True. Containment is Not Specified.]
 geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation	[Is static True. Containment is Not Specified.]
 historicalTerritory : Private Alias: temporal Territory Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDS reference: 3-02 mandatory for fixed facilities, optional for mobile facilities.	[Is static True. Containment is Not Specified.]
 id : Private Alias: 3-06 (primary) WSI	

ATTRIBUTES	
to prefixed with name space "facility:"	[Is static True. Containment is Not Specified.]
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links.	[Is static True. Containment is Not Specified.]
 title : Private Alias: 3-03 facilityName Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName	[Is static True. Containment is Not Specified.]
 type : Private Facility properties type; const facility.	[Is static True. Containment is Not Specified.]
 updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata.	[Is static True. Containment is Not Specified.]
 wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public groups (Class) facility Cardinality: [1..*]	Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]
 Association (direction: Unspecified) 9-01 responsibleParty Source: Public supervises (Class) facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1]	Target: Public has_geolocation (Class) historicalLocation Cardinality: [0..1]
 Association (direction: Unspecified) Source: Public (Class) facility	Target: Public (Class) historicalProgramAffiliation
 Association (direction: Unspecified) Source: Public (Class) facility Cardinality: [1]	Target: Public (Class) deployment Cardinality: [0..*]
 Association (direction: Unspecified) Source: Public describes (Class) facility Cardinality: [1]	Target: Public is_described_by (Class) environment Cardinality: [0..1]
 Association (direction: Unspecified) Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observation Cardinality: [1..*]

facilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

facilitySet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES	
 description : Private	[Is static True. Containment is Not Specified.]
 id : Private	[Is static True. Containment is Not Specified.]
 name : Private Alias: title	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [1..*]

Target: Public belongs_to (Class) facilitySet
Cardinality: [0..*]

historicalLocation

Class in package 'WMDR2-core'

historicalLocation
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-22
Alias temporalGeometry

ATTRIBUTES

 coordinates : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static True. Containment is Not Specified.]

 dates : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.

[Is static True. Containment is Not Specified.]

 methods : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static True. Containment is Not Specified.]

 type : str Private = MovingPoint

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public has_geolocation (Class)
historicalLocation
Cardinality: [0..1]

historicalObservingSchedule

Class in package 'WMDR2-core'

When are observations made? Describes a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".

Objects like

{"date": date, "schedule": str}, whereby "schedule" is a reference to a re-usable schedule object.

historicalObservingSchedule
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-22

ATTRIBUTES	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	
Date array aligned with observingSchedule.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) historicalObservingSchedule Cardinality: [1]	Target: Public uses (Class) schedule Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) observation Cardinality: [1]	Target: Public reports_with (Class) historicalObservingSchedule Cardinality: [0..*]

historicalProgramAffiliation

Class in package 'WMDR2-core'

historicalProgramAffiliation
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-22
Alias temporalProgramAffiliation

ATTRIBUTES	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	
Date array aligned with programAffiliation.	[Is static True. Containment is Not Specified.]
 programAffiliations : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	
Compact program-affiliation values.	[Is static True. Containment is Not Specified.]

ATTRIBUTES	
 reportingStatuses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) facility	Target: Public (Class) historicalProgramAffiliation

instrument

Class in package 'WMDR2-core'

Describes instrument capabilities, not a specific instrument instance.

For semantic consistency, terms like observedProperty and observedGeometry are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

instrument
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-05-22. Last modified 2026-06-13

ATTRIBUTES	
 5-09 manufacturer : Private	[Is static True. Containment is Not Specified.]
 5-09 model : Private	[Is static True. Containment is Not Specified.]
 id : Private	[Is static True. Containment is Not Specified.]
 observedGeometry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using http://codes.wmo.int/wmdr/Geometry	[Is static True. Containment is Not Specified.]
 observedProperties : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Specify as an array of values taken from http://codes.wmo.int/wmdr/ObservedVariable or as free text description.	[Is static True. Containment is Not Specified.]
 verticalRange : Private	

ATTRIBUTES

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) instrument
Cardinality: [0..1]

Target: Public belongs_to (Class) instrumentSet
Cardinality: [0..1]

 Association (direction: Unspecified) 5-09 instrument

Source: Public is_used_by (Class) observation
Cardinality: [1..*]

Target: Public uses (Class) instrument
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public (Class) deployment
Cardinality: [0..*]

Target: Public (Class) instrument
Cardinality: [1]

instrumentSet

Class in package 'WMDR2-core'

instrumentSet

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-22

ATTRIBUTES

 description : Private

[Is static True. Containment is Not Specified.]

 id : Private

[Is static True. Containment is Not Specified.]

 name : Private
Alias: title

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) instrument
Cardinality: [0..1]

Target: Public belongs_to (Class) instrumentSet
Cardinality: [0..1]

observation

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

observation
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES	
 contacts : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation-level contacts.	[Is static True. Containment is Not Specified.]
 exposure : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Compact exposure classification value.	[Is static True. Containment is Not Specified.]
 hisotricalSerialNumber : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [{"serialNumber": str, "date": date}, {..}]	[Is static True. Containment is Not Specified.]
 historicalDataPolicy : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Optional temporal data-policy history.	[Is static True. Containment is Not Specified.]
 historicalInstrumentOperatingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [{"instrumentOperatingStatus": str, "date": date}, {..}], whereby "instrumentOperatingStatus" values refer to http://codes.wmo.int/wmdr/InstrumentOperatingStatus	[Is static True. Containment is Not Specified.]
 historicalLocation : Private Alias: temporalGeometry Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 historicalOfficialStatus : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) [{"officialStatus": str, "date": date}, {..}], whereby "officialStatus" values refer to http://codes.wmo.int/wmdr/ObservationStatus	

ATTRIBUTES
[Is static True. Containment is Not Specified.]
◆ id : Private Observation identifier, e.g. observation:<observedVariable>. [Is static True. Containment is Not Specified.]
◆ keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords. [Is static True. Containment is Not Specified.]
◆ links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links. [Is static True. Containment is Not Specified.]
◆ observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry [Is static True. Containment is Not Specified.]
◆ observedProperty : Private Alias: OMS:ObservedProperty, iop:Variable The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2. NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific. Examples include: - observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll - observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text. - matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ... These elements will be supported by code lists from codes.wmo.int/wmdr2. Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc. [Is static True. Containment is Not Specified.]
◆ observingMethod : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) from (new) unified code list of observing methods, based on http://codes.wmo.int/wmdr/ObservingMethodAtmosphere http://codes.wmo.int/wmdr/ObservingMethodTerrestrial

ATTRIBUTES	
http://codes.wmo.int/wmdr/ObservingMethodEarth	[Is static True. Containment is Not Specified.]
 programAffiliation : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Program-affiliation timeline fragments inherited from WMDR1 observations.	[Is static True. Containment is Not Specified.]
 referenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Linked to heightAboveReferenceSurface, from http://codes.wmo.int/wmdr/ReferenceSurfaceType	[Is static True. Containment is Not Specified.]
 representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/Representativeness	[Is static True. Containment is Not Specified.]
 sourceOfObservation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Compact source-of-observation code value.	[Is static True. Containment is Not Specified.]
 time : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Observation temporal extent, date-resolution interval or null.	[Is static True. Containment is Not Specified.]
 title : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Human-readable title derived from domain, geometry, observed variable, and optional label.	[Is static True. Containment is Not Specified.]
 verticalDistanceFromReferenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) implement as an array of objects: [{ "value": float, "uom": str }, { ..., ... }], whereby "uom" from http://codes.wmo.int/wmdr/unit	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) 7-03 Temporal reporting period	
Source: Public describes (Class) observation	Target: Public uses (Class) reporting

ASSOCIATIONS	
Cardinality: [1]	Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public describes (Class) observation Cardinality: [1..*]	Target: Public uses (Class) procedure Cardinality: [0..1] ObservingProcedure(s)
 Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public (Class) observation Cardinality: [1]	Target: Public (Class) temporalGeometry Cardinality: [0..1]
describes	has_geolocatino
 Association (direction: Unspecified)	
Source: Public (Class) observation	Target: Public (Class) domain
 Association (direction: Unspecified) 5-09 instrument	
Source: Public is_used_by (Class) observation Cardinality: [1..*]	Target: Public uses (Class) instrument Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public groups (Class) observation Cardinality: [1..*]	Target: Public belongs_to (Class) observationSet Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) observation Cardinality: [1]	Target: Public reports_with (Class) historicalObservingSchedule Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observation Cardinality: [1..*]

observationSet

Class in package 'WMDR2-core'

grouping of observations (which are themselves time series)

observationSet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-22

ATTRIBUTES	
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Deployment temporal extent, date-resolution interval or null.	[Is static True. Containment is Not Specified.]
 id : Private Deployment identifier; source XML id is preserved when available.	[Is static True. Containment is Not Specified.]
 name : Private	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public groups (Class) observation Cardinality: [1..*]	Target: Public belongs_to (Class) observationSet Cardinality: [0..*]

procedure

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

procedure
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-02

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public describes (Class) observation Cardinality: [1..*]	Target: Public uses (Class) procedure Cardinality: [0..1] ObservingProcedure(s)

reporting

Class in package 'WMDR2-core'

Observation reporting metadata represented as aligned arrays. historicalTimeliness is itself a temporal wrapper.

reporting
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-22

ATTRIBUTES	
◆ dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of compact data-format values.	[Is static True. Containment is Not Specified.]
◆ dataPolicy : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of data-policy objects, preserving attribution when present.	[Is static True. Containment is Not Specified.]
◆ historicalAggregation : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reporting/aggregation periods, renamed from temporalReportingInterval.	[Is static True. Containment is Not Specified.]
◆ historicalTimeliness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional wrapper with timeliness durations and dates.	[Is static True. Containment is Not Specified.]
◆ internationalExchange : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of international-exchange flags.	[Is static True. Containment is Not Specified.]
◆ levelOfData : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of compact level-of-data values.	[Is static True. Containment is Not Specified.]
◆ numberOfObservationsInReportingInterval : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array copied from WMDR reporting metadata when present.	[Is static True. Containment is Not Specified.]
◆ referenceDatum : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reference-datum objects.	[Is static True. Containment is Not Specified.]
◆ referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Aligned array of reference-time-source values.	[Is static True. Containment is Not Specified.]
 timeStampMeaning : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of timestamp-meaning values.	[Is static True. Containment is Not Specified.]
 uom : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of compact unit code values.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) 7-03 Temporal reporting period Source: Public describes (Class) observation Cardinality: [1]	
Target: Public uses (Class) reporting Cardinality: [0..1]	

schedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

schedule
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 @type : Private JSCalendar object type; const Event.	[Is static True. Containment is Not Specified.]
 duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Occurrence duration describing coverage window or sampling duration.	[Is static True. Containment is Not Specified.]
 recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence override patches; null override values are invalid.	[Is static True. Containment is Not Specified.]

ATTRIBUTES	
 recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence rules. [Is static True. Containment is Not Specified.]	
 start : Private Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start. [Is static True. Containment is Not Specified.]	
 timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default. [Is static True. Containment is Not Specified.]	
 uid : Private JSCalendar-safe schedule uid. [Is static True. Containment is Not Specified.]	
 wmo.int:aggregation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific aggregation metadata. [Is static True. Containment is Not Specified.]	
 wmo.int:sampling : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific sampling metadata preserved from dataGeneration.sampling. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public describes (Class) historicalObservingSchedule Cardinality: [1] Target: Public uses (Class) schedule Cardinality: [1..*]	

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES

 **coordinates** : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static True. Containment is Not Specified.]

 **dates** : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.

[Is static True. Containment is Not Specified.]

 **methods** : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public (Class) observation
Cardinality: [1]

Target: Public (Class) temporalGeometry
Cardinality: [0..1]

describes

has_geolocatino